

The values listed below are general specs applying for PRECI-DIP spring-loaded connectors. Please see individual catalog page for additional and product specific technical data.

OPERATING TEMPERATURE RANGE	-55 ... +125 °C (-55 °... +85 °C with music wire spring)
CLIMATIC CATEGORY (IEC)	55/85/21
OPERATING HUMIDITY RANGE	Annual mean 75%
MAX. WORKING VOLTAGE	100 V _{RMS} /150 V _{DC} (2.54 mm grid)



PRECI-DIP connectors designated « Series Preci-Dip Connector » are recognized by UL LLC and listed under « Connectors for Use in Data, Signal, Control and Power Applications », File Nr. E174442.

MECHANICAL CHARACTERISTICS

	BASIC	LOW RESISTANCE	LOW RESISTANCE
	HOLLOW PISTON TYPE	SHAPED PISTON TYPE	CLIP IN-LINE DESIGN TYPE
MIN. INITIAL HEIGHT	3 mm	6 mm	10 mm
STROKE/HEIGHT RATIO	Max. 0.3	Max. 0.2	Max. 0.15
MAX STROKE	2 mm	2 mm	1.5 mm
MIN. INITIAL SPRING FORCE	0.2 N	0.2 N	0.2 N

ELECTRICAL CHARACTERISTICS

INSULATION RESISTANCE BETWEEN ANY TWO ADJACENT CONTACTS	Min. 10'000 MΩ at 500 V _{AC}
CAPACITANCE BETWEEN ANY TWO ADJACENT CONTACTS	Max. 1 pF

ENVIRONMENTAL CHARACTERISTICS

The sockets withstand the following environmental tests without mechanical and electrical defects:

- Dry heat steady state IEC 60512-11-9.11i / 60068-2-2.Bb: 125 °C, 16 h
 - Damp heat cyclic IEC 60512-11-12.11m / 60068-2-30.Db: 25/55 °C, 90 – 100 %rH, 1 cycle of 24 h
 - Cold steady state IEC 60512-11-10.11j / 60068-2-1.A: -55 °C, 2 h
 - Thermal shock IEC 60512-11-4.11d / 60068-2-14.Na: -55/125 °C, 5 cycles 30 min
 - Sinusoidal vibrations IEC 60512-6-4.6d / 60068-2-6.Fc: 10 to 500 Hz, 10 g, 1 octave/min, 10 cycles for each axis
 - Shock IEC 60512-6-3.6c / 60068-2-27.Ea: 50 g, 11 ms, 3 shocks in three axis
- During the above two tests, no contact interruption >50 ns does appear.

- Solderability J-STD-002A, Test A, 245 °C, 5 s, solder alloy SnAg3.8Cu0.7
- Resistance to soldering heat J-STD-020C, 260 °C, 20 s
- Moisture sensitivity J-STD-020C level 1



preci-dip

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SPRING-LOADED CONNECTORS

2.54 mm GRID / SINGLE ROW / DOUBLE ROW / SOLDER TAIL

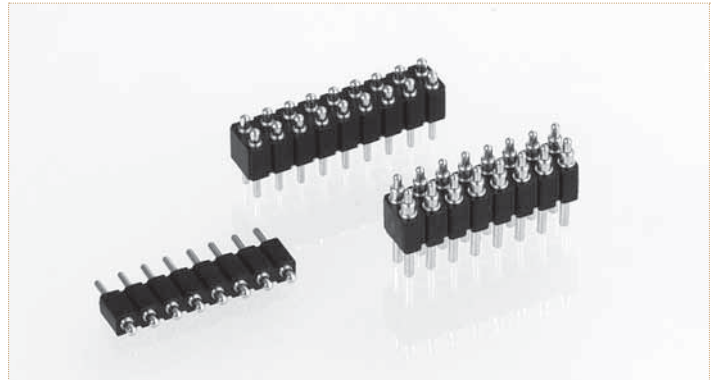
Low resistance modular connectors with spring-loaded contacts (SLC), solder tail. Contacts with improved, shaped piston design.

TECHNICAL SPECIFICATIONS (FOR GENERAL SPECS, SEE PAGE 31)

INSULATOR	Black glass filled polyester PCT-GF30-FR	
FLAMMABILITY	UL 94V-0	
PISTON AND BARREL	Brass CuZn36Pb3 (C36000)	
SPRING	Music wire DIN 17223, gold plated	
MAX. STROKE	1.4 mm	
FORCES	0.25 N initial	0.85 N at 1/2 stroke
MECHANICAL LIFE	50'000 cycles	
OPERATING CURRENT	Max. 3.5 A	
CONTACT RESISTANCE	10 mΩ (static measurement, halfway position)	

ORDERING INFORMATION ROHS COMPLIANT PARTS

PLATING CODE	BARREL	PISTON
S1	0.25 µm gold	0.5 µm gold
SS	0.5 µm gold	0.5 µm gold



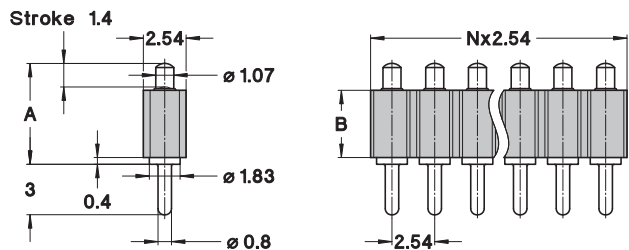
See page 178 for plating specs.

NNN number of poles: Replace **NNN** with the requested number of poles, e.g. 813-S1-**NNN**-10-014101 for a double row version with 16 pins becomes 813-S1-**016**-10-014101.

811-S1-**NNN**-10-**XXX**101

Straight SLC connector, low profile, solder tail, single row

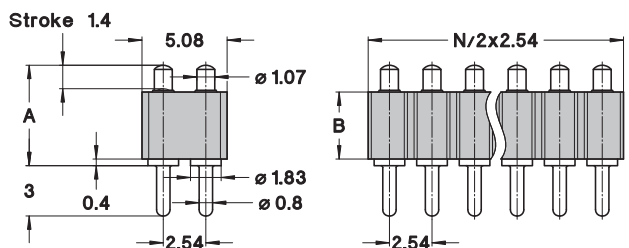
AVAILABILITY FROM 2 to 64 contacts
(standard number of contacts: 64)



813-S1-**NNN**-10-**XXX**101

Straight SLC connector, low profile, solder tail, double row

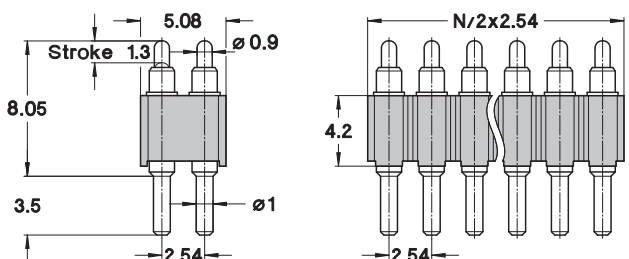
AVAILABILITY FROM 4 to 72 contacts
(standard number of contacts: 72)



813-SS-**NNN**-10-474101

Straight SLC connector, solder tail, double row

AVAILABILITY FROM 4 to 64 contacts
(standard number of contacts: 64)



AVAILABLE CONNECTOR HEIGHTS

INITIAL HEIGHT A (mm)	PLASTIC BODY HEIGHT B (mm)	XXX CODE
6	4	014
6.5	4	015
7	4	016
7.5	4	017